

EDUCATION

Bachelor's Degree Kwame Nkrumah University of Science and Technology Kumasi, Ghana
2021-2024 Bsc in Chemistry (First Class Honours)
- Class Rank: Top 3% of 230 students

PUBLICATIONS

1. Abakah, A., Manu, P*, Anfu, P.K*, Gyasi, K.F., **Okyere, H.**, Shardow, A.L.K., Osei-Poku, P., Kwarteng, A. (2025). **Immunoinformatics approach for developing a multi-epitope vaccine against *Pseudomonas aeruginosa* infection.** Medicine Advances (Wiley-VCH), *ID: MADV-2025-0096.R1. In review.*
2. **Okyere, H.**, Borquaye, L.S., Laryea, M.K. (2024). **Targeting *Staphylococcus aureus* AgrA Receptor: Computational Insights into Protein-Ligand Interactions and Protein-DNA Recognition.** **Thesis completed.**

SELECTED PORTFOLIO

3. **Okyere, H.** (2025). **Cyclistic bike-share data: uncovering rider patterns and health-centered marketing strategies.** [Video Presentation](#). Published on [Colab](#) and [GitHub](#).

RESEARCH EXPERIENCE

Research Assistant 2025	Vaccine Bionformatics KNUST (Ghana) Advisor: <i>Alexander Kwarteng Ph.D.</i> , Biochemistry Department Project Title: Immunoinformatics approach for developing a multi-epitope vaccine. • Used machine learning tools for epitopes prediction. • Performed molecular modeling and evaluated antigenicity for vaccine construction.
Undergraduate Research Student Full time Jan 2024-Aug 2024	Honours Thesis – Final Project, KNUST (Ghana) Advisors: <i>Michael K. Laryea Ph.D.</i> , <i>Lawrence S. Borquaye Ph.D.</i> Project Title: Quorum sensing inhibition: Targeting <i>S. aureus</i> AgrA Receptor. • Completed honour thesis on the development and application of GROningen Machine for Chemical Simulations (GROMACS) workflows for phosphorylated proteins and protein–DNA interactions, focusing on conformational changes in the AgrA protein receptor.
Undergraduate Research Student 20hrs/week Jan 2023-Aug 2023	Student Lead Investigator – Mini Project, KNUST (Ghana) Advisor: <i>Michael K. Laryea Ph.D.</i> Project Title: Natural product approaches to antimicrobial resistance (AMR) and quorum sensing. • Reviewed bacterial resistance and quorum sensing, proposing natural product-based strategies. • Presented weekly updates on research progress and drafted final-year project proposal.
Research Trainee 15hrs/week Sept 2023- Jan 2024	Virginia Tech University - Remote (USA) Advisor: <i>Luke Achenie Ph.D.</i> Project Title: Coarse-Graining and Molecular Simulations of Biomolecules • Learned coarse-graining of a membrane protein homodimer using Visual Molecular Dynamics. • Ran simulations with Nanoscale Molecular Dynamics (NAMD).
Research Trainee 15hrs/week 2024-present	Artificial Intelligence and Machine Learning (KNUST) <i>Student Research Initiative</i> • Applied machine learning to predict epitopes and active sites in bacterial regulatory proteins to support drug design. • Train peers on applied AI/ML workflows in computational biology.
Summer Research Student Full time Sept-Nov 2022	Department of Chemistry, KNUST (Ghana) Advisor: <i>Lawrence S. Borquaye, Ph.D.</i> • Studied phytochemicals in <i>Salacia debilis</i>, <i>Faurea speciosa</i>, and <i>Cassia donosa</i> for antimicrobial and antioxidant potential through experimental work. • Conducted molecular docking to explore binding properties of selected compounds.

TEACHING EXPERIENCE

Teaching Assistant Department of Chemistry, KNUST (Ghana)

Full time Delivered lectures and tutorials in analytical and inorganic chemistry to 400+ undergraduates.
2024- 2025 Supervised lab sessions, ensured adherence to safety and practical standards.
Mentored final-year students and guided project teams.

Track Lead Google Data Analytics Professional Certificates

2025 Cohort *Mentor Me Collective x Grow with Google*

Hosted weekly sessions, gave individualized feedback on [coding](#), [data cleaning](#), and modeling approaches to strengthen learning outcomes.

Used flashcards and targeted exercises to reinforce key concepts such as [Python scripting](#), [R](#), visualizations, and SQL queries.

Paired scholars to complete projects and ensured successful collaboration.

SELECTED ORAL & POSTER PRESENTATIONS

1. **Ghana Science Association Conference, CSIR Kumasi.** *Highlighted phosphorylation-driven conformational changes in AgrA-DNA binding.*

- Presented a [scientific poster](#) in 15-minute oral presentation and public session.

2. **Undergraduate Research Fest;** College of Science, Department of Chemistry, KNUST.

- Presented a [scientific poster](#) on my **final year project** in 10-minute oral presentation and public session.

3. **KNUST Scientific Research Conference,** Kumasi.

Okose, C.*, [Okyere, H.](#)*, Manu, P., Nketia, C., Borquaye, L.S. (2023). **Exploring the antifungal potential of *Salacia debilis* against *Candida albicans*.** **equal contribution*

- Presented a [scientific poster](#) in 15-minute oral presentation and public session.

FELLOWSHIPS & AWARDS

Teaching Teaching Assistant Award (KNUST)

award • Nominated by the Ghana Students' Chemical Society (GSCS), KNUST for exceptional undergraduate 2025 teaching and mentorship.

Academic award Second Best Female Chemistry Undergraduate (KNUST)

2024 • Ranked second among graduating female chemistry students (informal recognition).

Leaders Aspire Leaders Program (Aspire Institute – Harvard founded)

fellowship • Awarded a competitive three-month fellowship to participate in cross-cultural professional 2025 development and leadership training.

Healthcare Health Entrepreneurship Finalist (Mastercard Foundation – KNUST)

fellowship • Developed and pitched a diabetic-friendly health product business idea to a panel, handling market 2023 research, product concept design, and strategy presentation; gained experience in entrepreneurship, [teamwork](#), and [problem-solving](#).

Lab Research recognition

Best Research Undergraduate, Borquaye Lab (KNUST)

- Recognized for pioneering computational protocols and original research on protein regulation.

2024

Academic award University Scholar (KNUST Bursary)

2022 • Awarded this merit-based honor for academic excellence as a first-generation student with limited educational resources.

Completion Google Data Analytics Professional certificates | Kaggle badges (Coursera)

award • Awarded for completing 100+ hours of data science and machine learning training, acquiring skills in 2025 [Python](#), [R](#), [SQL](#), [data analysis](#), [visualization](#), and [predictive modeling](#).

SKILLS

Selected Methodologies:

- Molecular dynamics simulations, structural modelling, coarse-graining modeling, Immunology and vaccine modeling
- Data analysis, machine learning, predictive modeling, data cleaning, data visualization

Selected Laboratory Skills:

- TLC, FTIR, MS (GC/LC), NMR, UV-Vis, HPLC, microbial assays, antioxidant assays

Selected Tools & Platforms:

- AlphaFold, Spartan, NAMD/VMD, Schrödinger – Maestro, CHARMM GUI, GROMACS
- Tableau, GitHub, Google Colab, Kaggle, Linux

SELECTED OUTREACH ACTIVITIES

Participant **Stanford Biology PhD Preview (Stanford University)**

- 2025 • Selected through a competitive, faculty-nominated process to participate in an intensive graduate preparation program in the sciences.

Co-founder **Toproot Organics Ltd**

- Dec 2023-present* • Conduct research on natural ingredients and formulation strategies to develop and optimize diabetic-friendly sauces.

Consultant **Data Scientist, Nexyla IT Solutions**

- 2025 • Led the Data and AI team, planned the project roadmap, applied analytics and machine learning models to client dataset

Co-organizer **Girls in STEM & Science Mentorship**

- 2022 • Organized outreach sessions for girls in science and mentored senior high and undergraduate students in computational and experimental chemistry, promoting diversity in STEM.

Volunteer **Scholarly Support Activities**

- 2022-2023 • Supported undergraduate students in preparing for the College of Science departmental quiz teams, helping them develop knowledge, problem-solving skills, and teamwork.

Student **WiSTEM Participation, KNUST**

- Representative** • Selected among the top students to represent Yaa Asantewaa Girls' SHS at Women in Science, Technology, Engineering, and Mathematics (WiSTEM) events.
- Sept 2019* • Contributed to outreach, mentorship, and interactive science demonstrations for schools and community audiences.